

Pik Doctor: offline crop-leaf doctor for Maharashtra

Challenge 4: Small AI for Development · Agriculture · Marathi / Hindi / English · works with no internet

1. Challenge tackled

Farmers growing tomato, grape, maize, potato and bell pepper often find a sick leaf where there is no signal and no agriculture officer nearby. Pik Doctor diagnoses a leaf photo **entirely on the phone** and explains in Marathi, Hindi or English what it is, how urgent it is, what to do now and how to prevent it, with read-aloud. User: a smallholder farmer with an ordinary Android phone. After one visit with internet it works in airplane mode.

2. Tools / ML models used

- **Custom 0.75 M-parameter CNN** (4 conv blocks + 2 dense, 3 MB), trained **from scratch in pure NumPy** with hand-written, gradient-checked backprop (no PyTorch / pretrained weights were reachable in our build environment).
- **Plain-JavaScript inference engine** (no libraries). Verified against NumPy: max logit difference $2e-6$; image resize matches OpenCV INTER_AREA.
- **PWA + service worker** for installable, cache-first offline use; Web Speech API for read-aloud; OpenCV / NumPy for data; PlantVillage images (23 classes + a "not covered" class); Playwright for offline end-to-end tests.

3. What worked well

Held-out test (n = 994)	Result
Accuracy, 23 classes (crop not chosen)	94.7 %
Accuracy when farmer picks the crop	96.2 %
Accuracy on answers it commits to (confidence ≥ 70 %, 89 % of photos)	98.3 %
Diseased leaf called healthy	0.6 % (0.2 % with crop chosen)
Other-crop leaves refused as "not covered / not sure" (600 images)	97.5 %
Model size · speed	3 MB · ~ 0.4 s per diagnosis

- Calibrated confidence + a "not sure" threshold: the app prefers "retake the photo" to a wrong answer. Full flow tested in headless Chromium with the network switched off.
- Advice: 29 reusable snippets x 3 languages; no pesticide names or doses; always points to the KVK and Kisan Call Centre.

4. What was challenging

- **No ML stack.** Wrote the CNN, trainer and exporter in NumPy and a matching JS engine; proved they agree numerically.
- **Safety gap found in testing:** an apple leaf and a screenshot were called tomato diseases at ~ 90 % confidence. An unsupervised Mahalanobis guard flagged only 8 % of other-crop leaves, so we trained a real "not covered" class (other crops + synthetic non-leaf images): refusal rose to 97.5 %.
- **Accuracy:** first model 88 %, weak on tomato blight; a larger 96 px model with test-time augmentation reached 94.7 %.
- **Honest limits:** lab-style photos (field accuracy will be lower), near-duplicate images may inflate scores, Hindi/Marathi text not yet reviewed by a native-speaking extension worker.

5. How we spent our time

- Scoping and data: read the brief, chose crops, pulled 8,925 PlantVillage images · Model v1: NumPy CNN, 88 % · Engine + app: JS engine, trilingual UI, service worker · Model v2: 96 px, 94.7 % · Safety: found the out-of-scope failure, added the "not covered" class · Verification, README, videos, submission. (*Edit with your real hours.*)

If we had 24 more hours, we would...

collect real field photos from Maharashtra farms (also cotton, soybean, sugarcane, onion, pomegranate), have extension workers review every Marathi/Hindi sentence, and add on-device voice input.

Data: PlantVillage (Hughes & Salathé 2015; Mohanty et al. 2016). Code: see GitHub repository. Pik Doctor is a screening aid, not a diagnosis.